



Department Of Computer Science, CUI Lahore Campus

CSC102 - Discrete Structures

By

Mahwish Waqas

Lecture Outline

- Relations
 - What is a Relation?
 - Binary Relation
 - Representation of Relation
 - Relation Properties
 - Combining Relations

Relations

- The connections between people and things.
- Between people, family relation
 - 'to be brothers' x is a brother of y
 - 'to be older' x is older than y
 - 'to be parents' x and y are parents of z
- Between things, numerical relations
 - 'to be greater than' $x > y$ on the real numbers
 - 'to be divisible by' x is divisible by y on the set of integers
- Between things and people, legal relations
 - 'to be an owner' x is an owner of y

Cartesian Product

- The Cartesian product of sets A and B , denoted by $A \times B$, is the set of all ordered pairs of elements from A and B .
- $A \times B = \{(a, b) | a \in A \wedge b \in B\}$
- The elements of the Cartesian product are ordered pairs. In particular $(a, b) = (c, d) \leftrightarrow (a = c) \wedge (b = d)$
- $\{1, 2\} \times \{a, b\} = \{(1, a), (1, b), (2, a), (2, b)\}$
- $\{Mon, Tue\} \times \{Jan, Feb\} = \{(Mon, Jan), (Mon, Feb), (Tue, Jan), (Tue, Feb)\}$

Cartesian Product of More Than Two Sets

- Instead of ordered pairs we may consider ordered triples, or, more general, k-tuples
 - (a, b) , an ordered pair
 - (a, b, c) , an ordered triple
 - (a, b, c, d) , an ordered quadruple
 - $(a_1, a_2, \dots, a_k)$, a k-tuple
-
- Pairs, triples, quadruples, and k-tuples are elements of Cartesian products of 2, 3, 4, and k sets, respectively

Relations

- Relationship between elements of sets are represented using the structure called a relation, which is just a subset of the cartesian product of sets.

Binary Relation

- Let A and B be sets. A binary relation from set A to set B is any subset of $A \times B$.
- Binary relations represent relationship between two sets.

Example

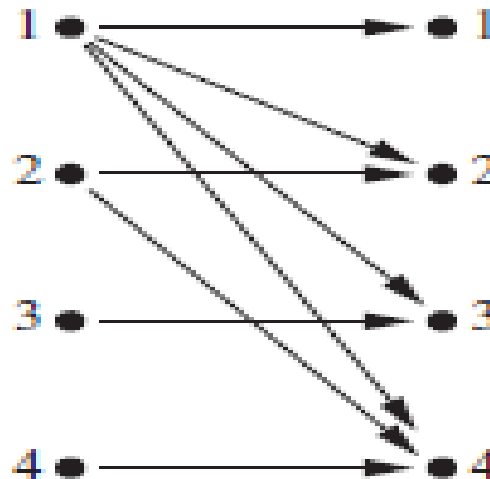
- $A = \{Ali, Junaid, Hassan\}$
- $B = \{CSC102, CSC222, CSC106\}$
- R : “*relation of students enrolled in courses*”
- $A \times B = \{(Ali, CSC102), (Ali, CSC222), (Ali, CSC106),$
 $(Juanid, CSC102), (Juanid, CSC222), (Juanid, CSC106),$
 $(Hassan, CSC102), (Hassan, CSC222), (Hassan, CSC106)\}$
- $R = \{(Junaid, CSC102), (Hassan, CSC222)\}$
- $(Junaid, CSC102) \in R$
- $(Ali, CSC102) \notin R$
- $(Junaid, CSC222) \notin R$
- $(Ali, CSC106) \notin R$

More Relations

- Relations can be generalized to subsets of cartesian products of more than two sets
- Any subset of the Cartesian product of 3 sets is called a ternary relation
 - 'x and y are parents of z' $\subseteq \text{People} \times \text{People} \times \text{People}$
- Any subset of the Cartesian product of k sets is called a k-ary relation

Functions as Relations

- A function $f:A \rightarrow B$ is a relation from A to B
- A relation from A to B is not always a function $f:A \rightarrow B$
- Relations are generalizations of functions!



Relations on a Set

- A (binary) relation from a set A to itself is called a relation on the set A .

$$A = \{1, 2, 3, 4\}$$

$$R_1 = \{(a, b) \mid a \text{ divides } b\}$$

$$A \times A = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

$$R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

$$R_1 \subseteq A \times A$$

Relations on a Set

- A (binary) relation from a set A to itself is called a relation on the set A .

$$A = \{1, 2, 3, 4\}$$

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_2 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_2 \subseteq A \times A$$

$$R_3 = \{(a, b) \mid a + b = 4\}$$

$$R_3 = \{(1, 3), (2, 2), (3, 1)\}$$

$$R_3 \subseteq A \times A$$

Describing Binary Relations

- A binary relation can be described using a list of pairs.

$$A = \{1, 2, 3, 4\}$$

$$R_1 = \{(a, b) \mid a \text{ divides } b\}$$

$$R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

Describing Binary Relations

- A binary relation can also be described using a matrix.
 - Rows are labeled with elements of A
 - Columns are labeled with elements of B .
- We write 1 in row a , column b if and only if $(a, b) \in R$; otherwise we write 0.

Describing Binary Relations

$$A = \{1, 2, 3, 4\}$$

$$R_1 = \{(a, b) \mid a \text{ divides } b\}$$

$$R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

$$M_{R_1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Describing Binary Relations

$$R_2 = \{(a, b) \mid a > b\}$$

$$R_2 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_2 \subseteq A \times A$$

$$M_{R_2} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Describing Binary Relations

- A binary relation can also be described using a directed graph.
- A graph of a relation $R \in A \times B$ consists of two sets of vertices labeled by elements of A and B .
- A vertex **a** is connected to a vertex **b** with an edge (arc) if and only if $(a, b) \in R$.

Reflexivity

- Note: from now on we will consider only binary relations on A .
- That is such relations are subsets of $A \times A$.
- Reflexivity: A binary relation $R \subseteq A \times A$ is said to be reflexive if $\forall a \in A, (a, a) \in R$
- $R = \{(a, b) | (a, b) \in \mathbb{Z} \times \mathbb{Z}, a \leq b\}$ is reflexive.

	a	b	c	d	e	...
a	1	.	.	.	.	.
b	.	1	.	.	.	.
c	.	.	1	.	.	.
d	.	.	.	1	.	.
e	.	.	.	.	1	.
...	.	.	.	.	.	1

Reflexivity

$$\forall_{a \in A} (a, a) \in R$$

$$A = \{1, 2, 3, 4\}$$

$$(1, 1), (2, 2), (3, 3), (4, 4)$$

$$R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

$$R_2 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$M_{R_1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Reflexive

If diagonal contains all 1's
then

Relation is **reflexive**.

else

Relation is **irreflexive**.

$$M_{R_2} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Irreflexive

Symmetry

- Symmetry: A binary relation $R \subseteq A \times A$ said to be symmetric if, $\forall a, b \in A, (a, b) \in R \rightarrow (b, a) \in R$
- $R = \{(a, b) | (a, b) \in \text{people} \times \text{people}, a \text{ and } b \text{ are siblings}\}$ is symmetric, because if a is a sibling of b, then b is a sibling of a

	a	b	c	d	e	...
a	0	1	0	0	0	.
b	1	0	0	1	0	.
c	0	0	0	0	1	.
d	0	1	0	0	0	.
e	0	0	1	0	0	.
...	.	.	.	.	.	.

Symmetry

$\forall_{a,b \in A}$, If $(a, b) \in R$ then $(b, a) \in R$

$$R_1 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$$

$$R_2 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$$

If $M_R = M_R^t$

then

Symmetric

$M_{R_1} \neq M_{R_1}^t$ *Not Symmetric*

else

$M_{R_2} \neq M_{R_2}^t$ *Not Symmetric*

Not Symmetric

Transitivity:

- Transitivity: A binary relation $R \subseteq A \times A$ said to be transitive if, $\forall a, b, c \in A, (a, b) \in R \wedge (b, c) \in R \rightarrow (a, c) \in R$
- $R = \{(a, b) | (a, b) \in \mathbb{Z} \times \mathbb{Z}, a \leq b\}$ is transitive, because if, $a \leq b$ and $b \leq c$ then $a \leq c$.

$\forall a, b, c \in A,$	If $(a, b) \in R \wedge (b, c) \in R$ then $(a, c) \in R$		Transitive
Case 1	True	True	True
Case 2	True	False	False
Case 3	False		True

Transitivity:

$$R_1 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M_{R_1} * M_{R_1} = M_{R_1}^2$$

M_R	M_R^2	Transitive
Non-Zero	Non-Zero	True
Non-Zero	Zero	True
Zero	Non-Zero	False

Anti-Symmetry

- Anti-Symmetry: A binary relation $R \subseteq A \times A$ said to be anti-symmetric, if $\forall a, b \in A, (a, b) \in R \wedge (b, a) \in R \rightarrow a = b$
- The relation Motherhood ('x is the mother of y') is anti-symmetric, because if x is a mother of y, then y is not the mother of x.

	a	b	c	d	e	...
a	0	0	0	0	0	.
b	1	0	0	0	0	.
c	0	0	1	0	0	.
d	0	0	0	0	0	.
e	0	0	1	0	0	.
...	.	.	.	.	.	.

Anti-Symmetry

$\forall a, b \in A$	<i>if $(a, b) \in R \wedge (b, a) \in R$ then $a = b$</i>		Anti-Symmetry
Case 1	True	True	True
Case 2	True	False	False
Case 3	False		True

$$M_{R_1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Anti – Symmetric

$$R_3 = \{(a, b) \mid a + b = 4\}$$

$$R_3 = \{(1, 3), (2, 2), (3, 1)\}$$

$(1, 3) \in R_3$ and $(3, 1) \in R_3$ but $1 \neq 3$

Not Anti – Symmetric

Equivalence Relations

- A relation on a set A is called an equivalence relation if it is reflexive, symmetric, and transitive.

Example

- Which of these relations on $\{0, 1, 2, 3\}$ are equivalence relations? Determine the properties of an equivalence relation that the others lack
- $\{ (0,0), (1,1), (2,2), (3,3) \}$
 - Has all the properties, thus, is an equivalence relation
- $\{ (0,0), (0,2), (2,0), (2,2), (2,3), (3,2), (3,3) \}$
 - Not reflexive: $(1,1)$ is missing
 - Not transitive: $(0,2)$ and $(2,3)$ are in the relation, but not $(0,3)$
- $\{ (0,0), (1,1), (1,2), (2,1), (2,2), (3,3) \}$
 - Has all the properties, thus, is an equivalence relation
- $\{ (0,0), (1,1), (1,3), (2,2), (2,3), (3,1), (3,2), (3,3) \}$
 - Not transitive: $(1,3)$ and $(3,2)$ are in the relation, but not $(1,2)$
- $\{ (0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,2), (3,3) \}$
 - Not symmetric: $(1,2)$ is present, but not $(2,1)$
 - Not transitive: $(2,0)$ and $(0,1)$ are in the relation, but not $(2,1)$

Combining Relations

- Relations from A to B are subsets of $A \times B$, two relations from A to B can be combined in any way two sets can be combined.
- Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$
- The relations $R_1 = \{(1,1), (2,2), (3,3)\}$ and
$$R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$$
- $R_1 \cup R_2$
- $R_1 \cap R_2$
- $R_1 - R_2$
- $R_2 - R_1$

Combining Relations

- Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$
- The relations $R_1 = \{(1,1), (2,2), (3,3)\}$ and
$$R_2 = \{(1,1), (1,2), (1,3), (1,4)\}$$
- $R_1 \cup R_2 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (3,3)\}$
- $R_1 \cap R_2 = \{(1,1)\}$
- $R_1 - R_2 = \{(2,2), (3,3)\}$
- $R_2 - R_1 = \{(1,2), (1,3), (1,4)\}$

Combining Relations

- Let R be a relation from a set A to a set B and S be a relation from B to a set C . The composite of R and S is the relation consisting of ordered pairs (a, c) , where $a \in A, c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.
- What is the composite of the relations R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?

Combining Relations

$$R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$$

$$S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$$

Find $S \circ R$.

If $(a, b) \in R \wedge (b, c) \in S$ then $(a, c) \in S \circ R$

$(1, 1) \in R \wedge (1, 0) \in S$ then $(1, 0) \in S \circ R$

$(1, 4) \in R \wedge (4, 1) \in S$ then $(1, 1) \in S \circ R$

$(2, 3) \in R \wedge (3, 1) \in S$ then $(2, 1) \in S \circ R$

$(2, 3) \in R \wedge (3, 2) \in S$ then $(2, 2) \in S \circ R$

$(3, 1) \in R \wedge (1, 0) \in S$ then $(3, 0) \in S \circ R$

$(3, 4) \in R \wedge (4, 1) \in S$ then $(3, 1) \in S \circ R$

Exercise

1. Find $R \circ R$. Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$
2. Find $S \circ R$ using matrix representing of the relations.
Let R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with
 $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and
 S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with
 $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$.

Exercise

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Examples

- Parenthood ('x is the parent of y')
- Brotherhood ('x is the brother of y')
- Neighborhood ('x is the neighbor of y')
- Ownership ('x is the owner of y')
- 'x divides y'
- $x \leq y$
- $x = y$
- $x < y$

Chapter Reading and Exercise

Chapter # 9

Topic # 9.1

Q-1,2,3,10,11-14,18-21,30,32

Topic # 9.3

Q-1,2,3,4,5,6,7,8,13-c,14,23-28